Casey Barton

Honolulu, HI · (808) 936-5654 · glacier.equilibrium@gmail.com · github.com/GlacierEQ · casey-barton-glaciereq.vercel.app

PROFILE

Forward-deployed AI systems engineer based in Honolulu. I embed with the teams whose workflows I'm deploying — scope the ambiguous process, build the narrow proof, harden what carries load — then feed what the deployment taught me back into the platform.

Three years running an evidence-grade agent control plane: GenAI/RAG/agent systems, evaluation and governance with explicit release thresholds, failure analysis with root-cause quarantine, MCP-style integrations, durable multi-agent workflows. Former home inspector — permits, site, materials: evidence → finding → remediation is how I work.

CORE COMPETENCIES

Engineering: Python · TypeScript · Node.js · SQL · Bash · REST APIs · webhooks · Supabase · Postgres · Vercel · GitHub Actions (CI/CD) · containers

AI & Agents: LLM agent frameworks · function calling · RAG · prompt & context engineering · MCP protocol · durable multi-agent workflows · gold sets & eval harnesses · LLM-as-judge

Governance & Reliability: evaluation with release thresholds · error taxonomies · failure analysis · expert adjudication & approval gates · policy enforcement · adversarial testing · observability/SLOs · rollback & recovery runbooks · audit receipts · idempotent automation

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Develop, configure, deploy, and optimize production GenAI/RAG/agent systems across a **1,398-repository** platform: agent governance, MCP-style integrations, context externalization, document workflows, infrastructure modeling.
- Evaluation gates every release: the document forge fails closed — any number not on the candidate kit-number allowlist raises **HonestyError** instead of rendering; the campaign engine composes assessments into **GO / NO-GO** with immutable evidence models and stable finding codes.
- Run evaluation harnesses with gold sets and LLM-as-judge (local Mistral API, live-verified) over retrieval, tool use, and multi-step agent behavior; error taxonomies and per-run failure analysis feed prompt and configuration tuning validated before anything ships.
- Integrated **65 MCP connectors** plus APIs, webhooks, and data pipelines — typed schemas, claim tracking, evidence boundaries, idempotent automation; every run emits audit receipts.
- Durable multi-agent workflows with typed state transitions and recoverable runs: planning with typed handoff contracts, memory across runs, error recovery with root-cause quarantine, human-in-the-loop escalation on consequential actions.
- Ship observability and recovery by default: structured logs, task IDs, correlation IDs, retry counts, success/failure metrics — planned / started / succeeded / failed / verified distinguished on every run; recovery restricted to contingency evidence declared before reassessment.

- Embed with the operators whose workflows I'm automating: demos and proof-of-concepts, technical requirements, milestones and implementation plans, end-to-end coordination — then harden what the deployment taught me into platform capabilities.

EDUCATION

University of Hawaii at Manoa 2016
Bachelor of Science

AWS Cloud Institute 2025
Cloud Computing program

Colorado State University
Undergraduate coursework

SELECTED PROOF

Job-App Helix — attributable campaign engine

- Composes assessments into campaign-level **GO / NO-GO** with immutable evidence models, stable finding codes, and fail-closed decisions.
- Recovery restricted to contingency evidence declared before reassessment — the engine never silently tunes values until they pass.

GlacierEQ estate — agent systems portfolio

- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate.
- Eval harnesses with gold sets and LLM-as-judge, adversarial review, and the fail-closed honesty gate — governance is engineered, not documented.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

AI / Agents: LLM agent frameworks · function calling · RAG · MCP protocol · durable multi-agent workflows · LLM evaluation · prompt & context engineering

Data & Infra: Supabase · Postgres · Vercel · GitHub Actions · REST APIs · webhooks · containers

Practice: demos & POCs · eval gates · error taxonomies · failure analysis · observability/SLOs · audit receipts · idempotent automation · runbooks

Evidence boundary: portfolio repositories are independent work. Domain-inspired demonstrations are not claims of company affiliation, production deployment, or certification. No pharma, biotech, or GxP delivery experience — evidence discipline (hash receipts, provenance envelopes, audit trails) is the transferable practice.